

Computer Graphics: Lecture 2

A brief History Real-time 3D

And what we're going to cover in class

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Welcome Back!

- Last lecture we learned the expectations of the class
- We learned about the importance of the GPU
- We saw briefly what we'd be learning [any reminders?]
- We learned which technologies we will use [can anyone give me a list?]
- We learned why it will be valuable to learn to program the GPU. It's more than just video games! (but the video games are cool too) [what other things can it do?]

This lecture:

We're going to learn context:

- We're going to see the unique requirements and motivations behind 3D graphics development, framed in their historical context.
- We're going to learn how 3D graphics turned into a radically different paradigm than ordinary software development.
- We're going to learn why that matters: the 3D graphics pipeline.

The pipeline

This pipeline concept is incredibly important.

We don't program 3D apps the same way as other apps. Everything you have learned on your computer science journey up to now will be useful, but you will need to go beyond basic synchronous programming.

But it seems very strange at first, so I want to give you some context about the field and some ancient history to motivate it.

Computer Graphics

As a field, computer graphics is the study of generating images with computers. That's it.

This seems like a very simple definition, and it is, but the field is anything but simple.

Humans who can see often use it as their primary sense: most of us are very visual creatures. Therefore, we are very demanding of the images that we look at.

Specifically, we are demanding in two ways...

What we demand of computer graphics

1. We demand that the images have certain qualities (realism, artistic style, clarity, etc.)
2. We demand that they be generated quickly.

Let's talk first about the qualities we want...

Kinds of generated images

There are many applications for computer graphics. The application determines what qualities we want the generated images to have:

- Scientific visualization: the key is on expressing mathematical concepts to humans quickly. So images need to be clear and understandable.
- Simulation: realism is important here.
- Art (including games): creative expression is important

These are jobs that exist. Someone has to program the graphics card to achieve the desired visual effects. Look for job titles such as “graphics engineer” or “technical artist”.

Computer Graphics is a Broad Term

Any image generated by a computer is technically “computer graphics”

- Your operating system’s desktop is computer graphics
- Photoshop is computer graphics
- A movie being played on a computer is computer graphics
- A 3D game is computer graphics

The latter two are especially important.



Games and Movies

There are basically two “modes” in which we generate 3D graphics:

- **Pre-rendered:** we know what images we will need in advance, such as the frames of a movie. We have time to **render** (draw) these images, so that when we're ready, they're easy to display on the screen. Obviously, faster is better, but it can take hours and still work.
- **Realtime:** we *don't* know the images we will need in advance. They depend on something, such as user input or randomness. This is the case for video games.

What do I mean by *not knowing* the images in advance?

A video game “frame”

Consider this game screenshot:



Questions?

In the beginning

One of the first video games

Then, a split

Way back in the old days, engineers did not even agree on **how a screen should work**.